A Simulation-Coupled Concurrent Engineering Approach to Next-Generation Low-Cost Counter-UAV Missiles

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Abstract The rapid proliferation of unmanned aerial vehicles (UAVs) and loitering munitions demands the development of cost-effective interception systems capable of ensuring reliable performance in dynamic operational environments. Project *SAGITTARIUS* introduces a simulation-coupled concurrent engineering framework for the preliminary design of a low-cost surface-to-air missile aimed at countering small hostile drones. The methodology integrates guidance, propulsion, aerodynamics, structures, and lethality within a unified workflow, enabling continuous exchange of boundary conditions and performance constraints across subsystems.

Trajectory simulations are employed to validate the missile performance against representative mission scenarios, while warhead lethality and structural analyses are iteratively harmonized with aerodynamic and propulsion requirements to maintain design coherence. A reference case involving a target UAV flying at 60 m/s, positioned 8000 m from the defended zone and 5000 m in altitude, was used to assess nominal behaviour and off-design robustness.

Results demonstrate that the integrated workflow achieves consistent performance and high kill probability across different operational profiles, highlighting the benefits of multidisciplinary coupling in early-stage design. The proposed approach emphasizes the importance of scenario-driven, simulation-based interaction between subsystems as a key enabler for affordable and adaptable Counter-UAV missile solutions.

Introduction

The rapid proliferation of unmanned aerial vehicles (UAVs) and loitering munitions has reshaped modern warfare, introducing new paradigms in surveillance, reconnaissance, and tactical engagement. Their low cost, agility, and swarm capability have increased the complexity of airspace protection, as small UAVs can operate at low altitude with limited radar visibility, posing asymmetric threats to both military and civilian infrastructures [1, 2].

Conventional defense systems, conceived for high-value targets, are economically unsustainable against large numbers of inexpensive drones. Consequently, research has focused on affordable short-range missile systems capable of ensuring reliable interception performance [3]. These systems must guarantee adaptability to diverse UAV classes and flight profiles, requiring advanced detection, guidance, and control strategies.

Addressing these challenges demands an integrated approach combining aerodynamic, structural, guidance, and propulsion design under a unified optimization framework. Modern missile development therefore relies on simulation-driven, multidisciplinary environments that enable concurrent progress across domains, accelerating design and improving overall coherence [4].

Within this context, *Project SAGITTARIUS* was developed at the University of Naples Federico II, in collaboration with MBDA Roma, as a next-generation surface-to-air missile optimized for Counter-UAV missions. The project applies a simulation-coupled concurrent engineering framework in which all primary subsystems—guidance, propulsion, aerodynamics, structures, and lethality—are developed in parallel through iterative data exchange and mission-driven constraints.

This paper outlines the design rationale and integrated workflow of the *SAGITTARIUS* project, demonstrating how scenario-based simulations and subsystem interaction lead to a coherent and adaptable missile configuration.

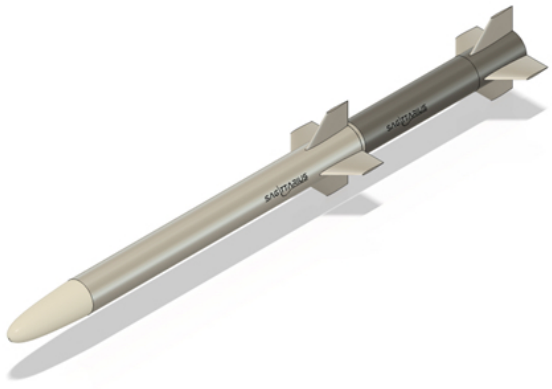


Figure 1: *SAGITTARIUS missile design and subsystem interaction concept.*

Methodology

The development of the *SAGITTARIUS* missile followed a concurrent engineering workflow aimed at ensuring tight integration among all subsystems from the earliest design phases. Each discipline was developed in parallel, with continuous data exchange and validation loops driven by mission-level simulations. The overall workflow is illustrated in Figure 2.

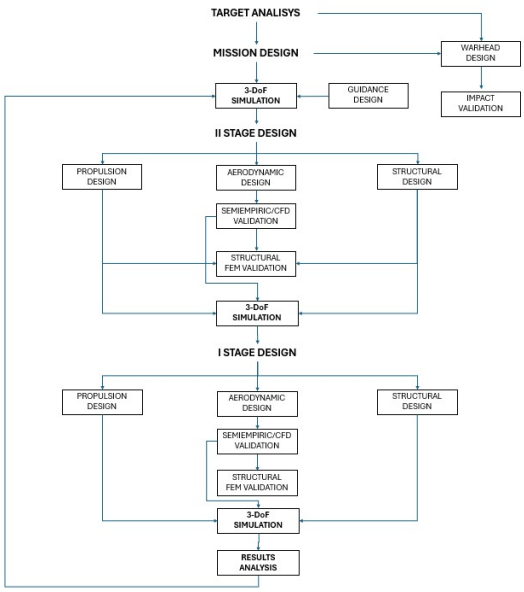


Figure 2: *Concurrent engineering workflow adopted for the SAGITTARIUS missile design.*

The process began with a comprehensive target analysis, defining the physical and operational characteristics of hostile UAVs, such as velocity range, altitude, and representative geometry. These parameters guided the mission design, establishing engagement requirements including an interception range of 8 km, target altitude of 5 km, and velocity of 60 m/s.

A 3-Degree-of-Freedom (3-DoF) trajectory simulation was employed to assess mission feasibility and derive key metrics—acceleration limits, flight path, and kill probability (P_k). The simulation outputs served as boundary conditions for subsystem design, while the guidance and control logic were developed in parallel using a Proportional Navigation (PN) scheme integrated into the model.

The second-stage design followed a concurrent approach, combining propulsion, aerodynamic, and structural development. Propulsion parameters such as thrust and burn time were refined through iterative feedback with trajectory results. Aerodynamics were assessed using semi-empirical and CFD-based analyses, whereas FEM simulations validated the structural response of major components. Each iteration of the 3-DoF model ensured alignment between subsystem performance and mission objectives.

After the second stage was finalized, its output conditions—velocity, attitude, and altitude—were transferred as inputs for the first-stage design, optimized to provide adequate thrust and stability during launch while maintaining overall mission coherence.

Missile Design & Reference Scenario

The interceptor was conceived as a compact, two-stage subsonic missile optimized for short-range air defence against small UAVs and loitering munitions. The architecture couples a booster stage for initial acceleration with a sustain stage for cruise and terminal engagement, balancing mass distribution, propulsion performance, and aerodynamic stability. The configuration was defined through a concurrent engineering process harmonizing subsystem constraints derived from trajectory simulations.

Table 1: Principal characteristics (II stage / I stage / Total).

Characteristic	II Stage (Sustain)	I Stage (Booster)	Total (Missile)
Total mass	11.224 kg	5.470 kg	16.694 kg
Propellant mass	2.524 kg	1.288 kg	3.812 kg
Diameter	0.0806 m	0.0806 m	0.0806 m
Length	0.976 m	0.456 m	1.432 m
Reference area S_{ref}	0.005 m^2	0.005 m^2	0.005 m^2
Nominal thrust	150 N	3626.5 N	—
Average burn time	30.0 s	0.8201 s	—

The design workflow employs a hybrid MATLAB/Simulink environment coupling a 3-Degree-of-Freedom (3-DoF) trajectory solver with guidance, control, and propulsion modules. The booster provides initial conditions to the sustain stage, whose dynamics include aerodynamic forces, gravity, and actuator response. Launch geometry is computed through an *auto-aiming* routine solving the relative motion with gravity correction. Guidance follows a Proportional Navigation (PN) law generating commanded accelerations, converted into fin deflections by the autopilot. The simulation ends when the range-to-target R drops below 1 m or interception becomes unfeasible, providing data for subsystem sizing and kill probability (P_k) estimation.

The reference mission involves a small UAV flying at constant altitude and speed toward a protected area, representing a typical Counter-UAV engagement. The target travels at 60 m/s and 5000 m altitude, starting 8000 m from the defended zone on a 180° inbound heading. The protected area radius is 3 km, defining the interception boundary for neutralizing the threat. Figure 3 shows the nominal engagement, highlighting missile and target trajectories together with the corresponding kill probability (P_k) evolution, confirming full interception and stable guidance under nominal conditions.

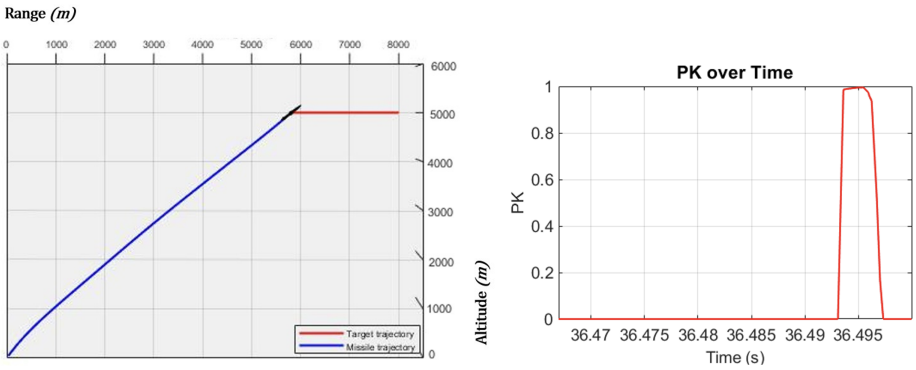


Figure 3: *Nominal engagement scenario showing missile and target trajectories with time evolution of kill probability (P_k).*

Off-Design Conditions

To assess the robustness of the *SAGITTARIUS* missile, several off-design simulations were conducted by varying the target geometry and initial conditions, while keeping control laws and propulsion settings identical to the nominal case.

This allowed the evaluation of guidance adaptability and overall system performance under realistic deviations from design conditions.Here two of these cases are represented.

The first configuration reproduces a *shallow dive attack*, with the target descending at 30° from 5000 m altitude and 8000 m ground distance at 60 m/s (Figure 4). The interceptor adapts its trajectory through a progressive climb, completing the engagement after about 35 s with stable guidance and nearly unitary lethality ($P_k \approx 1$). The second case simulates a *close-range engagement*, with the UAV starting 3000 m from the launch point at 5500 m altitude (Figure 5). Despite higher manoeuvre demand and shorter flight time (∼33 s), the missile maintains control stability and achieves $P_k \approx 0.9$, confirming adequate energy and guidance margin.

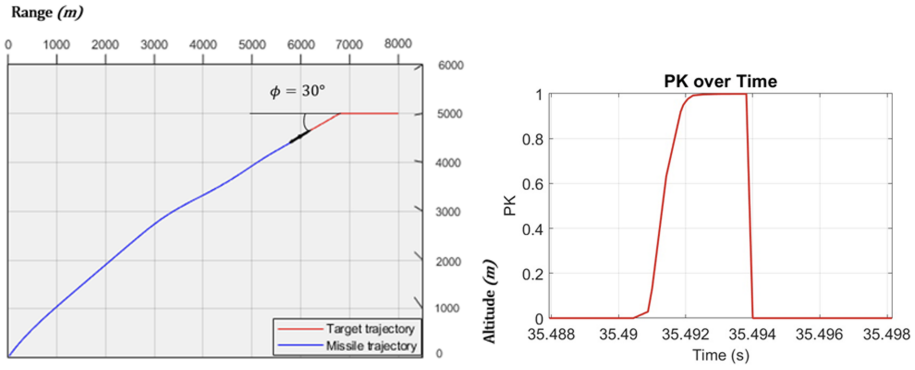


Figure 4: *Off-design scenario 1 – Shallow dive approach ($\phi = 30°$): missile and target trajectories with corresponding kill probability evolution.*

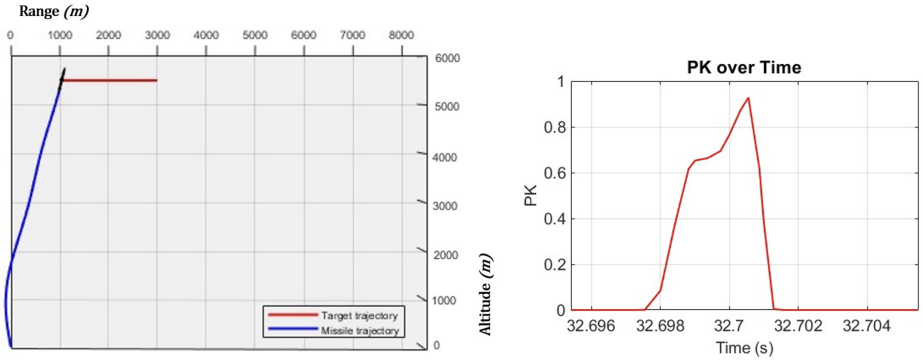


Figure 5: *Off-design scenario 2 – Close-range engagement: trajectory adaptation and time evolution of kill probability (P_k).*

Conclusion

The *SAGITTARIUS* validated a simulation-driven, concurrent engineering framework for the preliminary design of Counter-UAV missile systems. The integrated workflow ensured coherence among propulsion, aerodynamics, structures, guidance, and lethality, maintaining performance consistency across nominal and off-design scenarios.

Results confirmed stable guidance, aerodynamic efficiency, and high lethality under varied mission geometries, demonstrating the robustness of the proposed approach. The methodology proved effective in enabling reliable and adaptable missile configurations through early multidisciplinary coupling.

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